Tech4Bharat 2026 · Agriculture and Food Processing · Problem Statement 6.1

KisanSetu

An agentic AI-powered agricultural market access and intelligent selling network

From harvest to better markets



Growing the produce is only half the challenge. Getting a fair return is the other half

Six constraints act together. Fixing one alone rarely changes what the farmer earns.

A better price at another mandi is no use without affordable transport and a buyer there.

1 Information asymmetry

Net prices across markets, and how they are shifting, are hard to see.

2 Fragmented supply

Individual quantities are too small to negotiate with larger buyers.

3 Limited buyer access

Dependence on a narrow set of local buyers and intermediaries.

4 Transportation economics

A higher price can vanish after transport and handling costs.

5 Time pressure

Perishable produce leaves little time to search for better options.

6 Transaction uncertainty

Quantity disputes, quality deductions and delayed payments erode trust.

Farmers have access to information. What they need is coordinated action

Conventional information tools		KisanSetu: one shared agentic workflow
Display market prices	→	**Compare expected net realisation**
Show buyer listings	→	**Match available supply with buyer demand**
Provide individual recommendations	→	**Coordinate collective selling**
Show transport information	→	**Coordinate transport for confirmed sales**
Display market trends	→	**Help evaluate selling opportunities**
Provide transaction records	→	**Monitor the complete transaction lifecycle**

One intelligent network, from anticipated harvest to completed sale

Farmers register upcoming harvests, discover market opportunities, connect with verified buyers, combine produce with other farmers, arrange transport and manage transactions through one coordinated network of AI agents.

Eight services. One workflow.

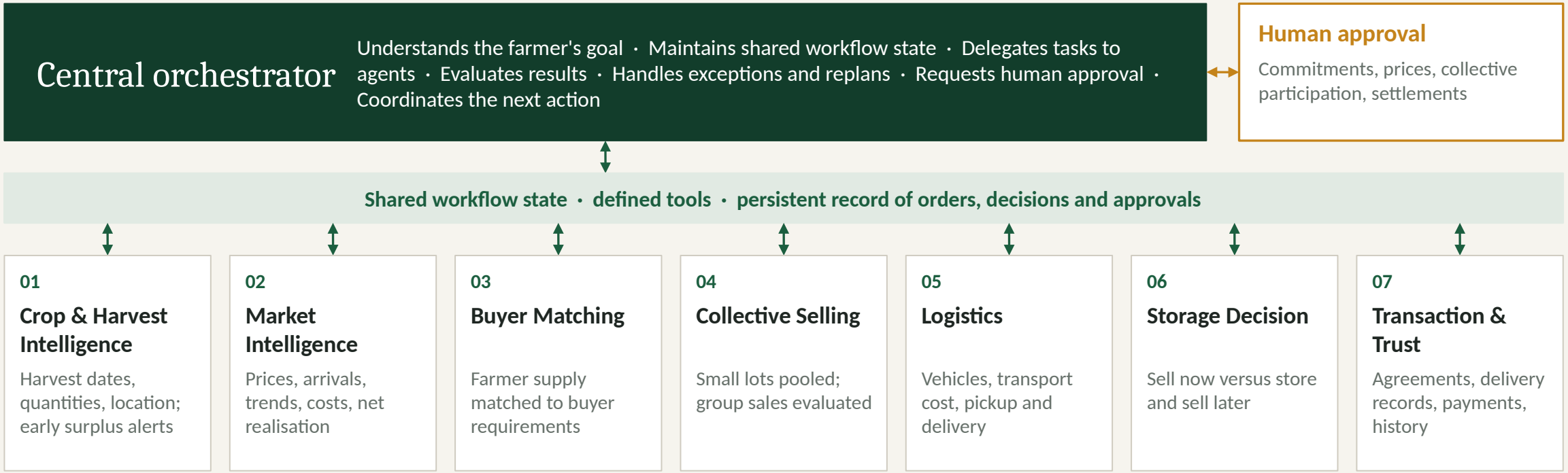
Each is a step in the same selling journey. None is a separate app.

The farmer says: crop, quantity, harvest date, location

1 **Crop and harvest intelligence** What, how much, when and where	2 **Market intelligence** Net realisation across markets	3 **Buyer matching** Supply matched to buyer demand	4 **Collective selling** Small lots pooled into bulk supply
5 **Logistics coordination** Shared, costed transport	6 **Storage decisions** Sell now, or store and sell later	7 **Transaction and trust** Records through to reconciliation	8 **Voice-first assistance** Multilingual, for low digital literacy

One shared harvest record → a farmer-approved sale, tracked to payment

Not another chatbot. A network of agents that plans, coordinates and acts



Language model handles intent understanding, planning, tool selection, coordination, exception handling and natural-language interaction.

Deterministic code handles price calculations, cost estimation, matching constraints, aggregation and optimisation.

One farmer, one harvest, one coordinated selling journey

Illustrative scenario. A farmer in Gujarat expects to harvest 3,000 kg of tomatoes in five days.



Harvest registration
Crop, quantity, date, location and quality, by voice

Market investigation
Markets compared on estimated net realisation

Buyer discovery
A processor needs 5,000 kg in the delivery window

Collective supply
Nearby farmers or an FPO supply the other 2,000 kg

Logistics planning
Vehicle capacity, pickup points and delivery cost

Transaction proposal
Price, deductions, transport and expected proceeds

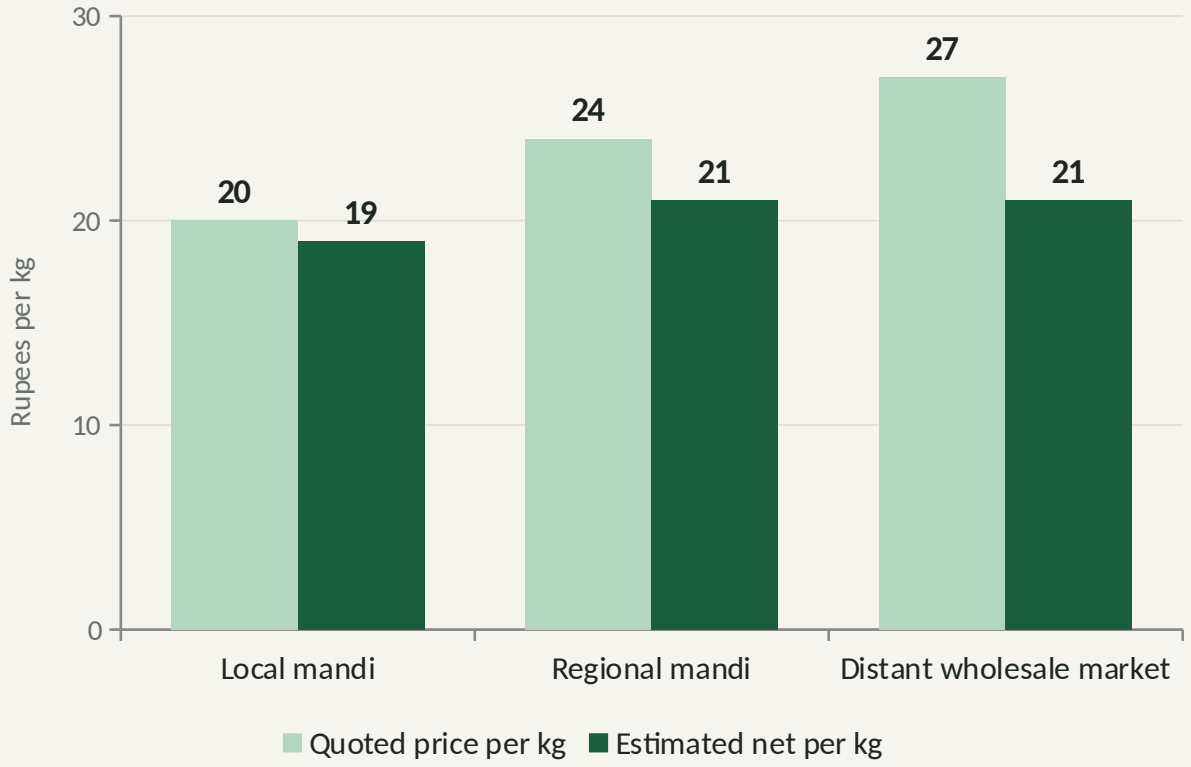
Farmer approval
Each farmer approves their own commitment

Execution and reconciliation
Dispatch, delivered quantity and payment tracked

When conditions change, the plan changes

A buyer withdraws, a transporter drops out or the harvest falls short. The orchestrator replans from the remaining options, and the farmers approve again. In the prototype, buyer demand is labelled real, sourced or simulated.

We optimise for what farmers actually earn, not just the price displayed on a screen



Illustrative figures, not market data.

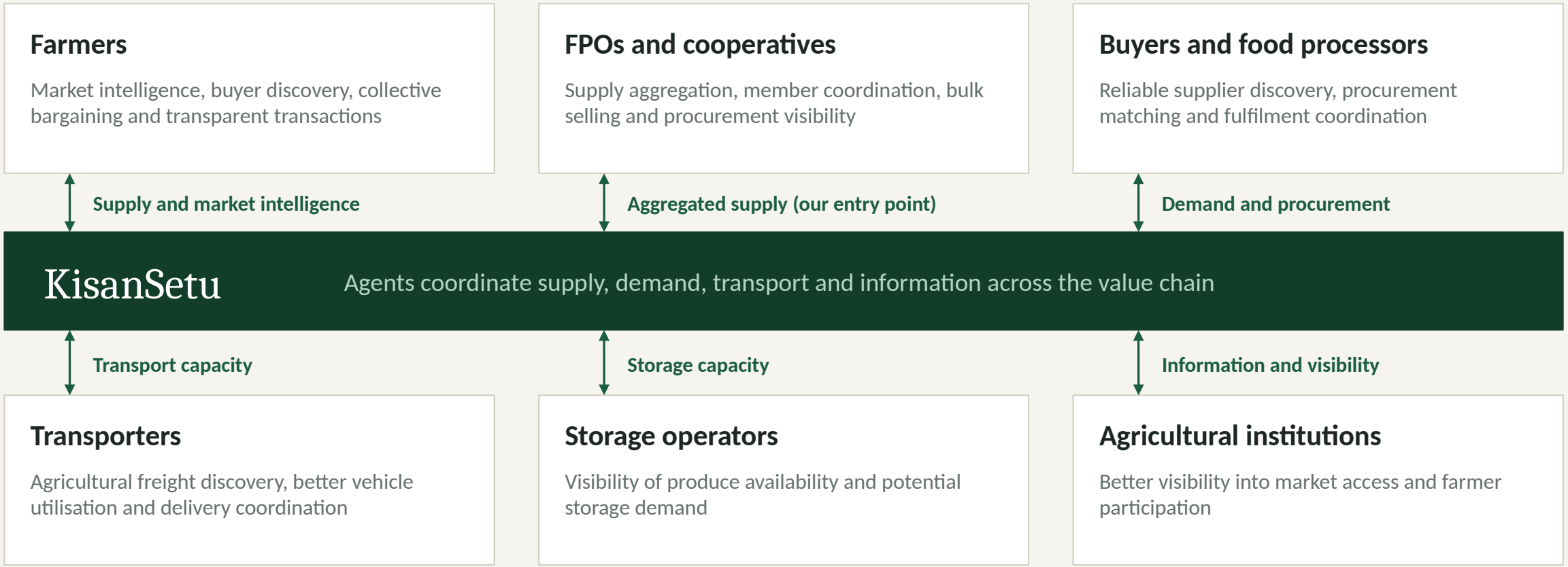
₹27 quoted. ₹21 earned.

The distant market quotes ₹7 per kg more than the local mandi, yet nets the same ₹21 as the regional mandi once transport and other costs are paid.

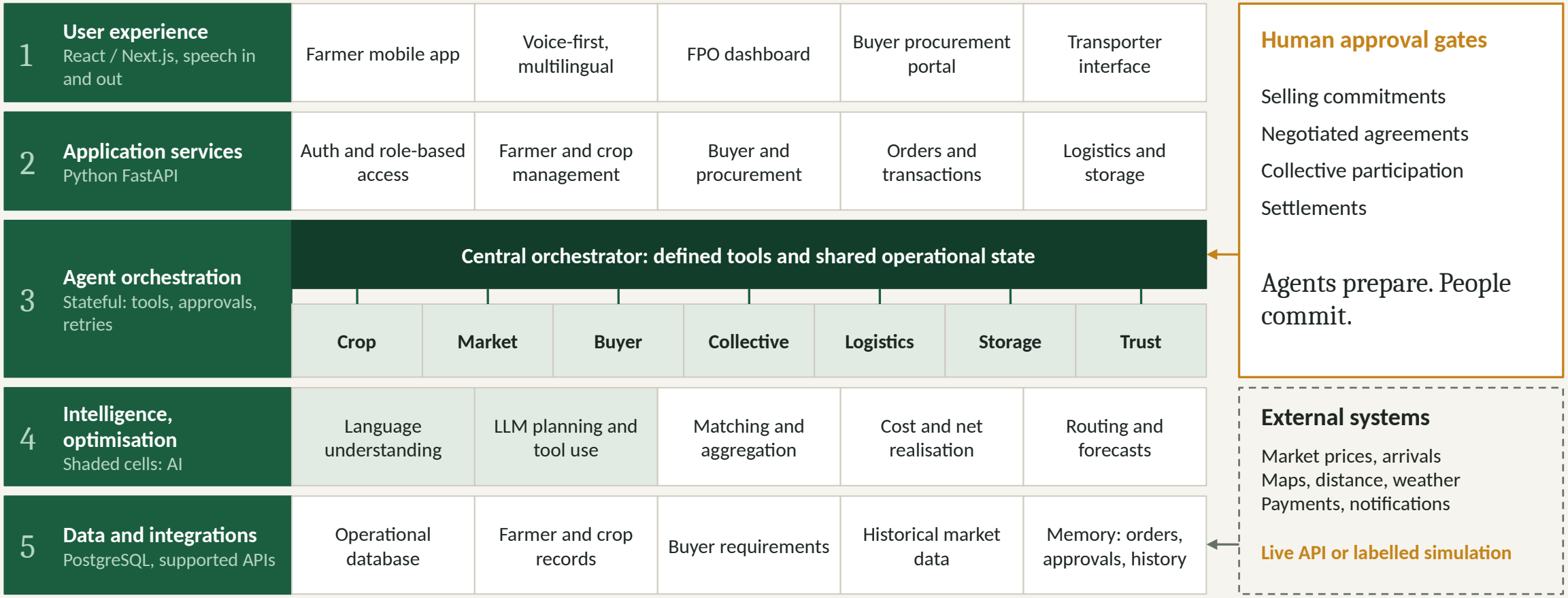
Estimated net for 1,000 kg	
Local mandi	**₹19,000**
Regional mandi	**₹21,000**
Distant wholesale market	**₹21,000**

KisanSetu ranks options by expected net, where cost data is available.

Empowering farmers while connecting the entire agricultural value chain



A coordinated agentic architecture built around shared state, reliable data and human control



Better market access can translate into stronger farmer economics

1 **Improved net realisation**
Compare actual economic returns, not headline prices

2 **Stronger bargaining power**
Aggregate supply collectively to reach larger buyers

3 **Reduced avoidable costs**
Better transport coordination and selling decisions

4 **Greater market transparency**
Clear records of offers, quantities, deductions and payments

5 **Improved market participation**
Small farmers reach procurement that is hard to access alone

How we will measure it

Target metrics, not results

1 Change in net realisation per kg

2 Farmers accessing new buyers

3 Volume of produce sold collectively

4 Transport cost per kg

5 Share of transactions completed

6 Average payment reconciliation time

7 Farmer repeat usage and retention

A farmer-first platform with multiple institutional revenue opportunities

Customer	Potential revenue model
FPOs and cooperatives	Subscription or institutional licence
Buyers and processors	Subscription: supplier discovery, procurement
Logistics providers	Transparent transaction or booking fees
NGOs and livelihood organisations	Institutional deployment, programme partnerships
Government and institutions	Licensing or programme-based deployment

Farmers do not pay to reach their own market

Essential market intelligence and basic selling assistance stay free or highly affordable for individual small farmers.

Scalability roadmap



1 One crop, one defined region

2 Validate with farmers and an FPO

3 Onboard local buyers and transporters

4 More crops, neighbouring markets

5 Extend through FPOs and institutions

6 Regional market-access networks

From fragmented agricultural supply to intelligent market access

Every farmer should have access to the market intelligence, buyers, bargaining power and coordination capabilities needed to obtain a fair and informed return for their produce.

KisanSetu: connecting every harvest to better opportunities

